

Homothety

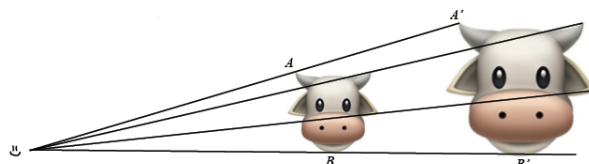
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1 Basic Properties

Given a point O and a number k different from 0 and 1, *homothety* with centre O and factor k is the transformation of the plane which sends point A to a point A' such that:

1. Points O , A and A' are collinear, and
2. $\overrightarrow{OA'} = k \cdot \overrightarrow{OA}$.



We denote such homothety by $\mathcal{H}(O, k)$.

The $\overrightarrow{OA'}$ and $\overrightarrow{OA}$ in the second part are directed segments, yet one can equivalently state that for $k > 0$ the rays OA and OA' coincide and for $k < 0$ they are mutually opposite. Regarding their different natures, it becomes natural to name $\mathcal{H}(O, k)$ a *positive homothety* if $k > 0$, and a *negative homothety* if $k < 0$.
p.s. Observe that choice $k = -1$ corresponds to point reflection.

2 Preservation of Ratios, Angles and Shapes

One of the basic application of homothety is the preservation of shapes, which seems trivial yet important. In particular, it may be difficult to prove the geometrical relation (e.g. concyclicity) of a handful of points.

2.1 Properties

Let $\mathcal{H}(O, k)$ be a homothety and denote the images of distinct non-collinear points A , B and C by A' , B' and C' respectively. Then:

1. Line $A'B'$ is *parallel* to AB . Moreover, $A'B' = k \cdot AB$.
2. Homothety *preserves angles and ratios*, i.e. $\angle A'B'C' = \angle ABC$ and $\frac{A'B'}{B'C'} = \frac{AB}{BC}$.

Therefore, *The image of a figure is a similar figure of the same orientation*. In particular:

1. The image of a line is a parallel line;
2. the image of a triangle is a similar triangle with corresponding sides parallel; and
3. the image of a circle is a circle.

2.2 Typical situations

Problem 1. Let $ABCD$ be a quadrilateral and $\odot$ be an arbitrary point not lying on the sides of $ABCD$. Prove that the centroids of $\triangle A\odot B$, $\triangle B\odot C$, $\triangle C\odot D$ and $\triangle D\odot A$ form a parallelogram.

Problem 2 (Nine-point circle). In a triangle ABC , let M_A, M_B, M_C be the three midpoints of the sides and H_A, H_B, H_C be the three foot of altitudes. Also, let A', B', C' be the midpoints of AH, BH, CH , where H is the orthocentre of $\triangle ABC$. Prove that $M_A, M_B, M_C, H_A, H_B, H_C, A', B'$ and C' lie on a circle with radius $\frac{R}{2}$ and centre being the midpoint of OH , where O and R are the circumcentre and circumradius of $\triangle ABC$.

2.3 Problems

Problem 3 (2006 Shortlist). Let $ABCD$ be a trapezoid with parallel sides $AB > CD$. Points K and L lie on the line segments AB and CD , respectively, so that $\frac{AK}{KB} = \frac{DL}{LC}$. Suppose that there are points P and Q on the line segment KL satisfying $\angle APB = \angle BCD$ and $\angle CQD = \angle ABC$. Prove that the points P, Q, B and C are concyclic.

Problem 4 (2013 China TST). The quadrilateral $ABCD$ is inscribed in circle ω . F is the intersection point of AC and BD . BA and CD meet at E . Let the projection of F on AB and CD be G and H , respectively. Let M and N be the midpoints of BC and EF , respectively. If the circumcircle of $\triangle MNG$ only meets segment BF at P , and the circumcircle of $\triangle MNH$ only meets segment CF at Q , prove that PQ is parallel to BC .

3 Collinearity and Concurrence I: Correspondence between Images

Intuitively, from the basic definition of homothety, one may prove collinearity by realising one of the desired points of is the centre of some homothety, and the remaining two points being homothetic. On the other hand, one may prove the desired point of concurrence being the centre of homothety to establish concurrence.

3.1 Existence of Homothety

1. Given two parallel segments AB and $A'B'$ of different length, *there exists a unique homothety* that maps A to A' and B to B' .
2. Let ABC and $A'B'C'$ be two non-congruent triangles with parallel corresponding sides. Then *there exists a unique homothety* that maps triangle ABC to triangle $A'B'C'$. As a result, *lines AA', BB' and CC' are concurrent*.

3.2 Typical situations

Problem 5 (Euler line). Let ABC be a triangle with circumcentre O , orthocentre H and centroid G . Prove that H, O and G are collinear with $\overline{HG} : \overline{GO} = 2 : 1$.

3.3 Problems

Problem 6 (2014 China TST). $ABCD$ is a cyclic quadrilateral with diagonals AC, BD perpendicular to each other. Let point E, F and G be on sides AB, BC and CD respectively, such that EF is parallel to AC and FG is parallel to BD . Let the projection of E onto CD be P , projection of F onto DA be Q and projection of G onto AB be R . Prove that QF bisects $\angle PQR$.

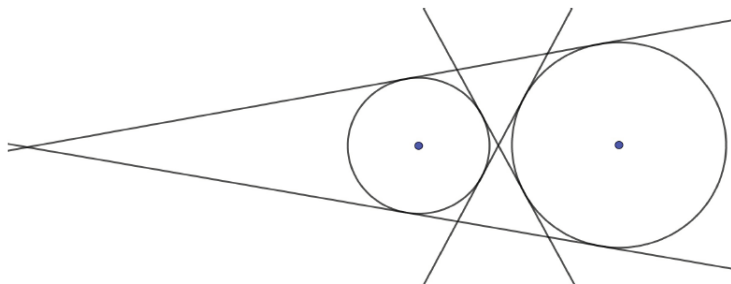
4 Circles and Tangents

Normally, it is difficult to deal with circle tangents, and homothety is a very powerful tool here, being a useful interpretation of mutually tangent circles or common tangent lines.

4.1 Homothety between two circles

Let ω_1, ω_2 be circles of different radii r_1, r_2 with centres O_1, O_2 respectively.

1. There exist *two* homotheties, one being *positive* (call it $\mathcal{H}^+$) and the other being *negative* (call it $\mathcal{H}^-$), that map ω_1 to ω_2 .
2. If common external tangents of ω_1 and ω_2 exist and intersect at $\odot^+$, then $\odot^+$ is the centre of homothety $\mathcal{H}^+$. Similarly, if common internal tangents of ω_1, ω_2 exist and intersect at $\odot^-$, then $\odot^-$ is the centre of homothety $\mathcal{H}^-$.
3. If ω_1 and ω_2 are internally tangent at $\odot^+$, then $\odot^+$ is the centre of $\mathcal{H}^+$. If they are tangent at $\odot^-$ externally, then $\odot^-$ is the centre of $\mathcal{H}^-$.



4.2 Typical situations

Problem 7. Circles ω_1, ω_2 are internally tangent at $\odot$. Chord AB of ω_1 is tangent to ω_2 at D . Show that $\odot D$ bisects $\angle A\odot B$.

Problem 8. Let t be a line. Circles ω_1, ω_2 , both lying on the same side of t , are tangent to it at T, U , respectively. Circle ω does not intersect t and is externally tangent to ω_1, ω_2 at K, L , respectively. Show that TK, UL , and ω pass through a common point.

Problem 9. Let ABC be a triangle and let its incircle ω and the A -excircle ω_a touch the side BC at D, E , respectively. Let K be the point on the incircle such that KD is a diameter. Then A, K, E lie on a single line.

4.3 Problems

Problem 10 (CGMO 2009). Circle Γ_1 , with radius r , is internally tangent to circle Γ_2 at S . Chord AB of Γ_2 is tangent to Γ_1 at C . Let M be the midpoint of arc AB (not containing S), and let N be the foot of the perpendicular from M to line AB . Prove that $AC \cdot CB = 2r \cdot MN$.

Problem 11 (CGMO 2013). Let $ABCD$ be a trapezoid with AB parallel to CD . The sides DA, AB, BC are tangent to circle ω_1 and AB touches ω_1 at P . The sides BC, CD, DA are tangent to ω_2 at Q . Prove that the lines AC, BD and PQ are concurrent.

Problem 12 (IMO 2006 Shortlist). Circles ω_1 and ω_2 with centres O_1 and O_2 are externally tangent at point D and internally tangent to a circle ω at points E and F respectively. Line t is the common tangent of ω_1 and ω_2 at D . Let AB be the diameter of ω perpendicular to t , so that A, E, O_1 are on the same side of t . Prove that lines AO_1, BO_2, EF and t are concurrent.

Problem 13 (IMO 2008). Let $ABCD$ be a convex quadrilateral with $BA \neq BC$. Denote the incircles of triangle ABC and ADC by ω_1 and ω_2 respectively. Suppose that there exists a circle k tangent to ray BA beyond A and to the ray BC beyond C , which is also tangent to the lines AD and CD . Prove that the common external tangents to k_1 and k_2 intersect on k .

5 Collinearity and Concurrence II: Multiple Homothety

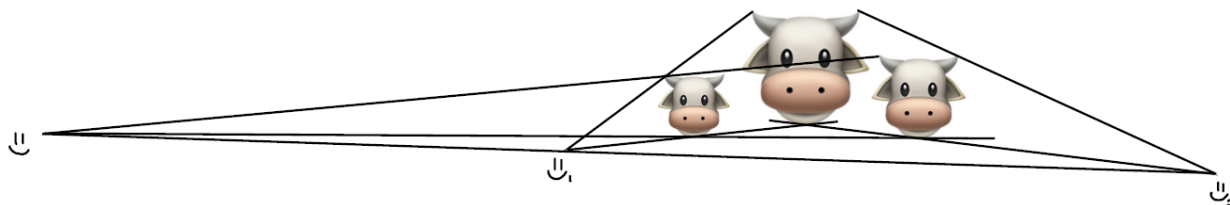
Besides one centre and two homothetic points, one can also prove collinearity by merely playing around homothety centres. By fully understanding and mastering the following property, one may manage to prove some insane-looking problems.

5.1 Composition of two Homotheties

Let $\mathcal{H}_1 (\sim_1, k_1)$ and $\mathcal{H}_2 (\sim_2, k_2)$ be homotheties such that $\sim_1 \neq \sim_2$ and $k_1 k_2 \neq 1$. Then:

1. Their composition (i.e. the transformation of the plane in which we perform $\mathcal{H}_1$ first and then apply $\mathcal{H}_2$ to the result) is *again a homothety*.
2. Moreover, the centre of the combined homothety *lies on the line* $\sim_1 \sim_2$.

Note that the factor of the combined homothety is $k_1 k_2$, which restricts the combined homothety being positive or negative.



5.2 Typical situations

Problem 14 (Monge's Theorem). Let ω_1, ω_2 and ω_3 be circles such that the common external tangents of ω_1 and ω_2 intersect at H_3 , those of ω_2 and ω_3 intersect at H_1 , and those of ω_3 and ω_1 intersect at H_2 . Then the points H_1, H_2 and H_3 are collinear.

5.3 Problems

Problem 15 (IMO 2020 Shortlist). Let $ABCD$ be a cyclic quadrilateral with no two sides parallel. Let K, L, M , and N be points lying on sides AB, BC, CD , and DA , respectively, such that $KLMN$ is a rhombus with $KL \parallel AC$ and $LM \parallel BD$. Let $\omega_1, \omega_2, \omega_3$, and ω_4 be the incircles of triangles ANK, BKL, CLM , and DMN , respectively. Prove that the internal common tangents to ω_1 and ω_3 and the internal common tangents to ω_2 and ω_4 are concurrent.

Problem 16 (CGMO 2011). The A -excircle Ω of $\triangle ABC$ touches BC at M . The points D, E lie on the sides AB, AC respectively such that DE is parallel to BC . The incircle ω of $\triangle ADE$ touches DE at N . Denote the centre of ω be I and that of Ω be J . If $BI \cap DJ = F$ and $CI \cap EJ = G$, prove that the midpoint of FG lies on MN .